

Data Mining Notes

Academic Year 2026/2027

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Chapter 1

Introduction to Data Mining and the KDD Process

1.1 Definition and Fundamental Motivations

The constant technological progress in data generation and storage has led to an exponential growth of available information, both in commercial and scientific contexts. The modern paradigm of data collection is often summarized in the principle of gathering any possible data, anywhere and anytime, in the expectation that such data may acquire value not only for the original purpose for which they were collected, but also for secondary purposes not initially foreseen.

Definition 1.1 (Data Mining). **Data Mining** is the application of efficient techniques and algorithms to the analysis of large data collections, aimed at the extraction of valid, novel, potentially useful, and interpretable patterns (hidden knowledge).

Data Mining does not consist in the simple execution of queries on traditional databases (for example, retrieving the average balance via SQL), but rather in the automatic or semi-automatic discovery of non-trivial structures, dependencies, or regularities in the data.

1.1.1 Application Perspectives

The motivations behind the adoption of Data Mining techniques can be distinguished primarily according to two points of view:

- **Commercial point of view:** Companies manage enormous volumes of data (*data warehousing*): web data (Yahoo, Facebook), e-commerce

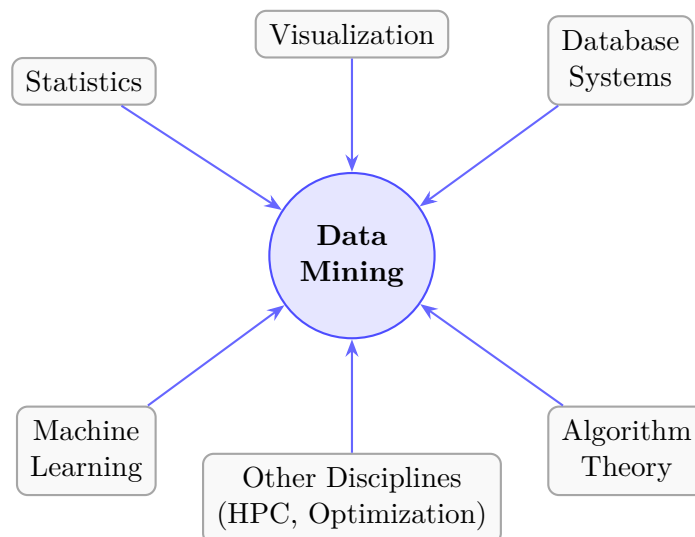
transactions and retail sales (Amazon), credit card records and bank accounts. In the face of increasingly cheap and accessible computing power, competitive pressure requires extracting value from such masses of data to improve and personalize the services offered (for example in *Customer Relationship Management* - CRM).

- **Scientific and social point of view:** In disciplines such as astronomy (*Sky Survey Data*), Earth observation (NASA EOSDIS satellites collect petabytes of environmental data per year), genomics (gene expression, proteomics), neuroimaging (fMRI), and complex numerical simulations, data are generated at such speeds as to make manual analysis impossible. Data Mining supports scientists in automatic analysis and in the guided formulation of new hypotheses.

Big Data represent to all intents and purposes a proxy of social life (purchasing behaviors, lifestyles, opinions, relationship networks, territorial mobility), offering great opportunities to address global issues: from the efficiency of healthcare to the search for clean energy sources, from forecasting climate impacts to optimizing agricultural production.

1.1.2 A Confluence of Disciplines

Data Mining is not an isolated discipline, but rather a synergistic confluence of several areas of computer science and applied mathematics:



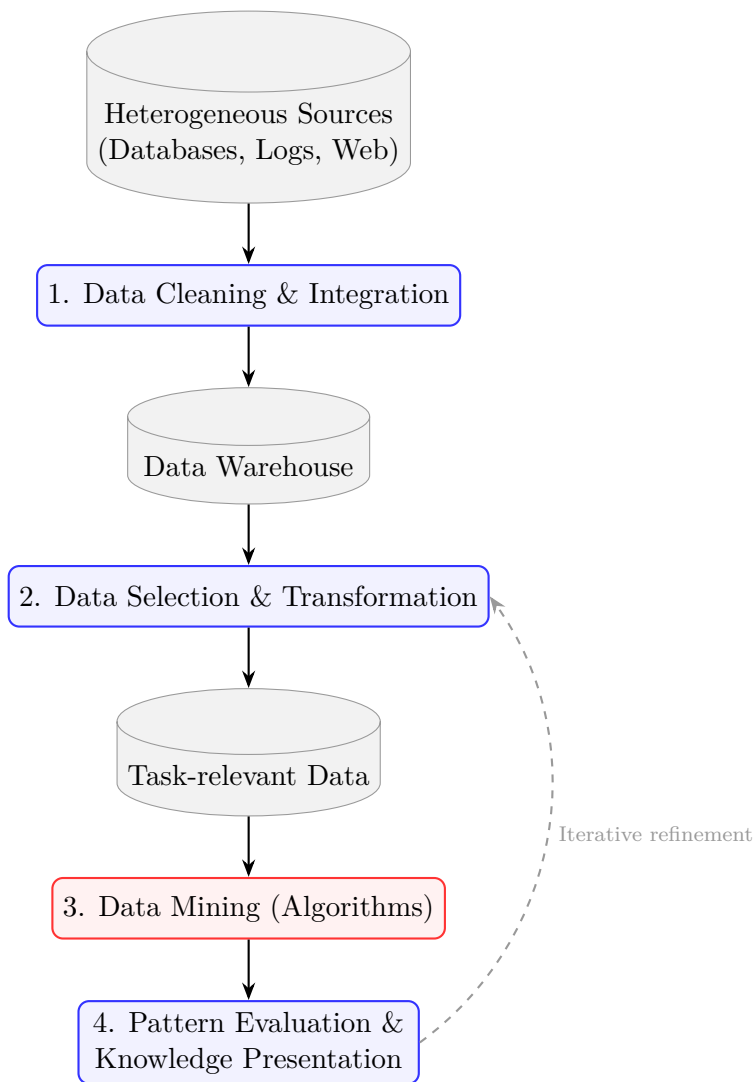
1.2 Types of Data Origin: Primary and Secondary

In setting up an analytical project, it is essential to distinguish the origin of the analyzed data:

- **Primary Data:** Original data collected specifically for a precise investigation objective. They have not undergone any preventive alterations or manipulations by third parties.
- **Secondary Data:** Data that had already been previously collected for operational purposes or different from the current one and made available to the analyst. They often derive from the merging of heterogeneous sources and require deep normalization and quality verification operations.

1.3 The KDD (Knowledge Discovery in Databases) Process

Data Mining constitutes a specific, although central, phase within the broader knowledge discovery process, known as **KDD** (*Knowledge Discovery in Databases*). The KDD is an iterative and interactive process composed of several sequential phases with feedback loops:



Each phase performs specific tasks:

1. **Business Understanding:** Understanding the domain objectives and translating the business problem into a formal data analysis problem.
2. **Data Cleaning and Data Integration:**
 - *Data Cleaning:* removal of noise, correction of syntactic errors, and handling of inconsistencies.
 - *Data Integration:* aggregation and reconciliation of data from multiple and heterogeneous sources, typically flowing into a **Data Warehouse** (a structured and optimized repository for analysis and decision support).

3. Data Selection and Data Transformation:

- *Data Selection*: extraction of the subset of attributes and records strictly relevant to the investigated task (*task-relevant data*).
- *Data Transformation*: transformation of features through aggregations, rescalings, normalizations, or discretizations so that they are suitable for the algorithm.

4. **Data Mining**: The actual execution of intelligent algorithms for the extraction of patterns and mathematical models.

5. **Pattern Evaluation and Knowledge Presentation**: Identification of truly significant patterns through objective and subjective metrics, elimination of redundancy, and graphical visualization of results using tools interpretable by the end user.

1.4 Taxonomy of Data Mining Tasks

Following the classification of Fayyad et al. [1], Data Mining tasks are divided into two macro-categories:

- **Prediction Methods**: Aim to estimate the future or unknown value of a specific target variable starting from the values of other observed variables (predictor attributes).
- **Description Methods**: Aim to identify human-readable and interpretable patterns that describe the intrinsic structure and hidden relationships in the data.

1.4.1 Predictive Tasks: Classification

Classification aims to learn an objective function f that maps each attribute record x into a specific discrete class label $y \in \mathcal{Y}$.

The dataset is divided into two disjoint sets:

- **Training Set**: Used to train the classifier.
- **Test Set**: Used to evaluate the generalization capability of the model on unseen instances during the learning phase.

Example 1.2 (Classification Applications). Among the main and most widespread application areas of classification models we find:

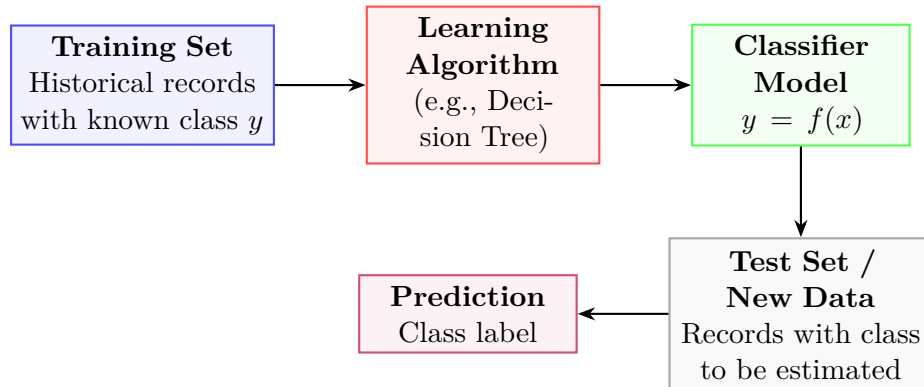


Figure 1.1: Conceptual workflow of the classification process.

- **Credit Worthiness Evaluation:** Predicting whether a loan applicant will repay the debt (*Yes/No*) based on employment, education level, and years of residence.
- **Bank Fraud Detection:** Classifying credit card transactions as legitimate or fraudulent. Features include amount, time, geographical location, and historical spending profile.
- **Churn Prediction:** Estimating the probability that a telecommunications customer will abandon the service for a competitor, analyzing call duration, data traffic volume, and recorded complaints.
- **Other applications:** Automatic categorization of news (sports, finance, politics), clinical diagnosis (benign or malignant tumor), land cover classification from satellite images.

1.4.2 Descriptive Tasks

1. Clustering

Clustering consists in organizing a set of objects into distinct groups (called *clusters*), such that:

1. The distance between objects belonging to the same cluster (**intra-cluster distance**) is **minimized** (maximum internal homogeneity).
2. The distance between objects belonging to different clusters (**inter-cluster distance**) is **maximized** (maximum external separation).

Clustering is an unsupervised technique: it does not require a priori classes.

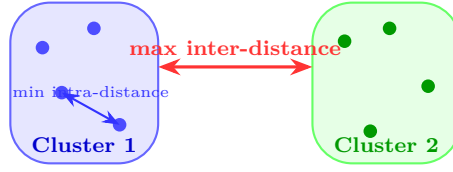


Figure 1.2: Fundamental geometric principle of clustering.

- **Market Segmentation:** Grouping customers based on socio-demographic data and purchasing habits to target differentiated marketing campaigns.
- **Document Clustering:** Grouping similar textual documents based on the joint frequencies of keywords (for example, the analysis of the Enron corporate email archive).
- **Summarization and Earth Sciences:** Reducing the complexity of massive spatial series by partitioning sea surface temperature (*Sea Surface Temperature* - SST) and primary production (*Net Primary Production* - NPP) measurements in the Northern and Southern hemispheres.

2. Association Analysis (Discovery of Association Rules)

Given a set of transactions, each consisting of a subset of items taken from a global set of items, association analysis aims to discover probabilistic dependencies in the form of rules:

$$X \implies Y \quad (1.1)$$

where X and Y are disjoint itemsets ($X \cap Y = \emptyset$). The rule indicates that the presence of the items in the antecedent X systematically predicts the contextual presence of the item or items in the consequent Y .

Example 1.3 (Market Basket Analysis). Considering a history of cash register receipts: The algorithm extracts rules with high statistical evidence, such as:

TID	Purchased Items
1	Bread, Coca-Cola, Milk
2	Beer, Bread
3	Beer, Coca-Cola, Diapers, Milk
4	Beer, Bread, Diapers, Milk
5	Coca-Cola, Diapers, Milk

$$\{\text{Milk}\} \implies \{\text{Coca-Cola}\}, \quad \{\text{Diapers, Milk}\} \implies \{\text{Beer}\}$$

Such information guides the strategic placement of goods on shelves and the design of cross-selling offers.

Other applications: Diagnosis of alarms in telecommunication networks (identifying temporal patterns of network alarms that trigger jointly to identify systemic root failures); medical informatics (correlation between combinations of clinical symptoms, blood tests, and emerging pathologies).

3. Anomaly / Deviation / Change Detection

It consists in identifying objects that deviate significantly from the average or expected behavior of the population. Anomalies can be generated by instrumental failures or entry errors (and in this case represent *noise* to be purged), or they can reflect exceptional events of extreme application interest (credit fraud, cyber intrusions against corporate networks, monitoring of sudden deforestation of forest cover via remote sensors).

1.5 Methodological and Computational Challenges in Data Mining

The application of traditional statistical analysis and database methods is often inadequate due to five primary challenges:

1. **Scalability:** Algorithms must handle data volumes on the order of terabytes or petabytes without computational complexity and memory footprint exploding.
2. **High Dimensionality:** The presence of hundreds or thousands of attributes exposes one to the so-called *curse of dimensionality*, where the geometric space becomes extremely empty and distances tend to lose discriminative power.
3. **Complex and Heterogeneous Data:** Need to process semi-structured or unstructured structures (graphs, texts, genomic sequences, spatiotemporal series).
4. **Data Ownership and Distribution:** Data are frequently geographically distributed across multiple organizations and subject to strict legal privacy and security constraints.

5. **Non-Traditional Analysis:** Need to extract knowledge from continuous streams in real-time (*data stream mining*) without the possibility of storing the entire dataset.

Chapter 2

Data Understanding and Data Exploration

2.1 Objectives of Data Understanding

The data understanding phase (*Data Understanding*) constitutes an indispensable prerequisite for any Data Mining activity. Before selecting or applying any algorithm, the analyst must acquire an overall view of the structure, distribution, and intrinsic quality of the dataset with respect to the project objectives.

In this preliminary phase, it is necessary to answer crucial methodological questions:

- What types of attributes compose the dataset?
- What is the overall level of data quality (presence of noise, errors, or duplicates)?
- How are the numerical and categorical variables distributed?
- Are there evident correlations or redundancies among the attributes?
- How are outliers structured and what application meaning do they assume?
- What strategies should be adopted for missing values?
- Is it necessary to derive or transform attributes to make mining more effective?

2.2 Types of Datasets

The collected data present themselves in different logical-geometric structures based on the nature of the application domain:

1. Record Data:

- **Data Matrix:** A set of objects described by the same fixed set of numerical attributes. A dataset of m objects and n attributes is formally represented by an $m \times n$ matrix:

$$\mathbf{X} = \begin{pmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,n} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m,1} & x_{m,2} & \cdots & x_{m,n} \end{pmatrix} \quad (2.1)$$

In this formalization, each object x_i is a point in an n -dimensional vector space $\mathbb{R}^n$.

- **Document Data (Bag-of-Words):** Each textual document is represented by a term vector (*term vector*). Each component of the vector corresponds to a word in the vocabulary, and the associated value indicates its frequency of occurrence in the text.
 - **Transaction Data:** Each record represents a transaction (identified by a TID) and contains a variable set of purchased items or observed events.
2. **Graph Data:** Structures composed of nodes and edges that model relationships and connectivity. Examples include the network of hyper-text links of the World Wide Web, social network graphs, and molecular structures (for example, the planar molecule of benzene C_6H_6).

3. Ordered Data:

- **Sequences of transactions:** Temporal successions of purchases or events for each user.
- **Genomic data:** Discrete biological sequences composed of nucleotide characters $\{\text{A}, \text{C}, \text{G}, \text{T}\}$.
- **Spatio-Temporal Data:** Continuous or sampled measurements over geographical coordinates and regular time intervals (for example, the monthly average temperature of the land and ocean surface).

2.3 Objects and Attributes

A dataset is a collection of **data objects** and their corresponding **attributes**.

- An **object** (also known as a record, point, case, sample, entity, or instance) represents a single entity of the domain of interest.
- An **attribute** (also known as a variable, field, characteristic, dimension, or feature) represents a measurable property or characteristic of an object (for example, age, taxable income, or eye color).

2.3.1 Measurement Scales of Attributes

The type of measurement scale of an attribute is defined by the set of algebraic properties and mathematical operators that preserve the meaning of the measurement:

1. **Distinctness** ($=, \neq$): Possibility to distinguish whether two values are equal or different.
2. **Order** ($<, >$): Possibility to establish an order relationship between values.
3. **Meaningful differences** ($+, -$): Presence of a constant unit of measurement that allows quantifying the interval between two values (zero is conventional/arbitrary).
4. **Meaningful ratios** ($*, /$): Presence of an **intrinsic absolute zero** (indicating the total absence of the measured quantity), which makes the proportional comparison between quantities meaningful.

2.3.2 Further Distinctions of Attributes

- **Binary Attributes:** Nominal attributes that exclusively admit two states (0 and 1). They are distinguished into:
 - *Symmetric Binary:* Both outcomes have equal informational importance (for example, biological gender).
 - *Asymmetric Binary:* One of the two outcomes is rare but of extreme informational relevance (for example, a positive outcome to a diagnostic test for a rare disease). By convention, the value 1 is associated with the event of greater interest.

Scale	Properties	Description	Examples	Statistical Operators
Nominal	Distinctness ($=, \neq$)	Qualitative values or categorical labels without intrinsic ordering	SSN, ZIP code, eye color, gender	Mode, entropy, χ^2 test, contingency
Ordinal	Distinctness and Order ($=, \neq, <, >$)	Values with meaningful ordering but non-constant interval	Educational degree, Likert scale (1–10), mineral hardness	Median, percentiles, rank correlation (Spearman)
Interval	Order and Differences ($+, -$)	Quantitative differences with meaning; zero is conventional	Temperature in $^{\circ}\text{C}$ or $^{\circ}\text{F}$, calendar dates	Mean, variance, standard deviation, Pearson, t and F tests
Ratio	All 4 properties ($=, <, +, *, /$)	Presence of absolute zero; ratios between physically valid measurements	Age, income, mass, counts, Kelvin	Geometric and harmonic mean, percentage variation

Table 2.1: Taxonomy of attribute measurement scales (Stevens).

- **Discrete vs Continuous Attributes:**

- *Discrete*: Assume a finite or countably infinite set of values (typically represented by integers, counts, or categories). Binary attributes constitute a special case of discrete attribute.
- *Continuous*: Assume values in a continuous subset of real numbers $\mathbb{R}$ (temperature, weight, coordinates). In computational practice, they are measured and stored using floating-point representations (*floating-point*) with finite precision.

2.4 Data Quality

Data quality is a critical factor for the success of any analytical initiative. As highlighted in Thomas Redman's studies, problems arising from poor quality data typically cost organizations between 10% and 20% of their overall revenues. If a credit risk model is trained on flawed or erroneous data, the system will reject solvent debtors and disburse loans to insolvent individuals.

2.4.1 Dimensions of Data Quality

Quality problems manifest themselves under various dimensions:

1. **Syntactic Accuracy:** The recorded value violates formal constraints or does not belong to the predetermined domain (for example, the string "female" or the presence of letters in a numeric field). Such anomalies can be identified automatically through simple schema checks.
2. **Semantic Accuracy:** The data is formally correct with respect to the type, but false in the reference reality (for example, associating the name "John Smith" with the female gender or recording an age of 180 years). Detecting these discrepancies requires domain rules (*business rules*) or cross-referencing with reliable external sources.
3. **Completeness:** Systematic lack of values for certain attributes or absence of entire records. A typical violation of completeness is **sampling bias** (for example, the analysis of insolvency risk carried out on a historical archive that only includes customers already approved by the bank, excluding a priori low-income profiles).

4. **Unbalanced Data:** Strongly asymmetric distribution of classes (for example, fraudulent transactions equal to 0.1% of the total or defective products on highly automated production lines).
5. **Timeliness:** Obsolete degree of data updating compared to the current dynamics of the observed phenomenon.
6. **Duplicate Data:** Presence of identical or nearly identical records relating to the same entity (for example, the same customer registered multiple times with different email addresses following the merger of two databases). Such discrepancies are addressed through deduplication and *record linkage* techniques.

2.5 Descriptive Statistics and Univariate Exploration

2.5.1 Measures of Central Tendency

- **Sample Mean ($\bar{x}$):** For a sample of size m :

$$\bar{x} = \frac{1}{m} \sum_{i=1}^m x_i \quad (2.2)$$

The mean is a distributive measure (it can be calculated by partitioning the dataset into subsets), but it is **extremely sensitive to the presence of outliers**.

- **Median:** The value that occupies the central position in the increasing order of the observations (or the mean of the two central values if m is even). It is a robust measure, with a breakdown point (*breakdown point*) of 50%.
- **Trimmed Mean:** Mean calculated by excluding a fixed percentage (for example 5% or 10%) of the most extreme values at both ends.
- **Mode:** The value or category that occurs with the maximum absolute frequency. A distribution can be unimodal, bimodal, multimodal, or modeless if all observations appear only once.

2.5.2 Skewness and Multimodality

- **Positively Skewed / Right-skewed:** The right tail of the distribution is elongated. Most observations are concentrated on values lower than the arithmetic mean (Mode < Median < Mean). A typical example is the distribution of **incomes**: the vast majority of citizens earn low or medium incomes, while a small number of individuals earn extremely high incomes, pulling the mean upwards.
- **Bimodal Distribution:** Presence of two distinct peaks in the frequency density. It typically signals the influence of an unstated binary latent variable (for example, the distribution of human body weight in a mixed population, influenced by the binary variable sex).

2.5.3 Measures of Dispersion

- **Range:** Difference between maximum value and minimum value:
Range = $x_{\max} - x_{\min}$.
- **Sample Variance (s^2) and Standard Deviation (s):**

$$s^2 = \frac{1}{m-1} \sum_{i=1}^m (x_i - \bar{x})^2, \quad s = \sqrt{s^2} \quad (2.3)$$

The standard deviation measures the dispersion around the mean; it is zero only if all observations are identical ($s = 0$), otherwise $s > 0$. It is also sensitive to outliers.

- **Robust Measures of Dispersion:**

- *Absolute Average Deviation* (AAD): $AAD = \frac{1}{m} \sum |x_i - \bar{x}|$.
- *Median Absolute Deviation* (MAD): $MAD = \text{median}(|x_i - \text{median}(x)|)$.
- *Interquartile Range* (IQR): Distance between the third quartile Q_3 (75th percentile) and the first quartile Q_1 (25th percentile):

$$IQR = Q_3 - Q_1 \quad (2.4)$$

2.5.4 Box Plot and Five-Number Summary

The **Box Plot** (box and whisker plot) offers a non-parametric visual summary of the distribution based on the **five-number summary**:

$$[\text{Minimum}, Q_1, \text{Median} (Q_2), Q_3, \text{Maximum}]$$

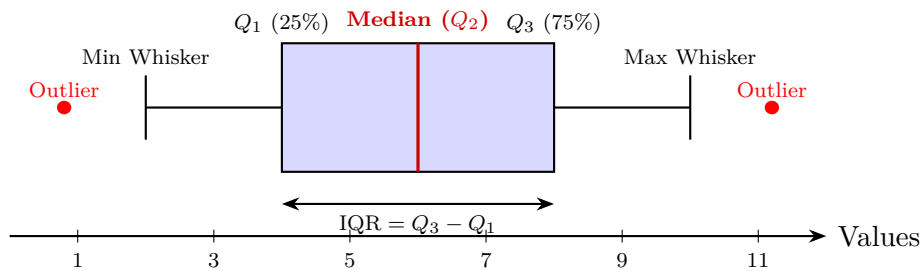


Figure 2.1: Geometric anatomy of a Box Plot with identification of outliers.

The limits of the whiskers are conventionally calculated by extending up to the most extreme observed value within the interval $[Q_1 - 1.5 \times IQR, Q_3 + 1.5 \times IQR]$. The observations that fall outside such thresholds are visually labeled as **outliers**.

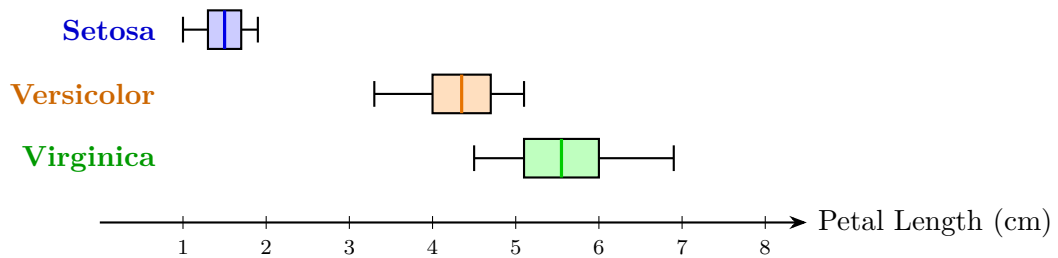


Figure 2.2: Example of Conditional Box Plot: distribution of the *Petal Length* attribute conditioned on the flower species in the Iris dataset.

Critical Comparison: Box Plot vs Histogram

Although the box plot effectively synthesizes position and dispersion, **it is not able to show the real shape of the distribution**. Two distributions having radically different densities (for example, a uniform unimodal distribution and a strongly bimodal distribution with an empty central valley) can produce identical values of Q_1 , median, Q_3 , Minimum, and Maximum, and therefore generate the exact same box plot. The histogram is essential to detect multimodality.

2.5.5 Bar Chart for Categorical Attributes

The **Bar Chart** visually represents the frequency distribution of a qualitative or categorical attribute. Each discrete category is represented by a vertical

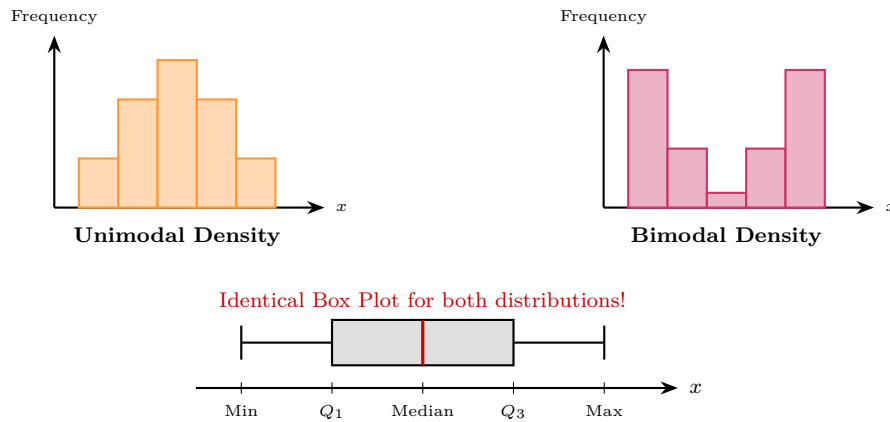


Figure 2.3: Critical comparison: two distributions with diametrically opposite behaviors (unimodal vs bimodal with empty center) generate the same five-number summary and the same Box Plot.

(or horizontal) bar, whose height corresponds to the absolute count or relative frequency. Unlike histograms, bars are separated by empty spaces to highlight that the modalities do not constitute a numerical continuum.

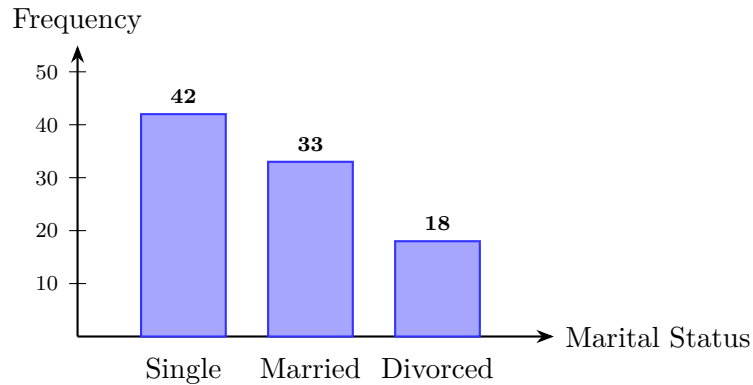


Figure 2.4: Example of a Bar Chart for a nominal categorical attribute (Marital Status).

2.5.6 Histograms for Numerical Attributes and Choice of Bins

A **histogram** shows the empirical frequency distribution of a continuous numerical attribute. The domain of variability of the quantity is partitioned

(*discretized*) into a predetermined number k of contiguous intervals called *bins* (or class widths). For each bin, the height of the overlying bar is proportional to the number of sample observations that fall within it.

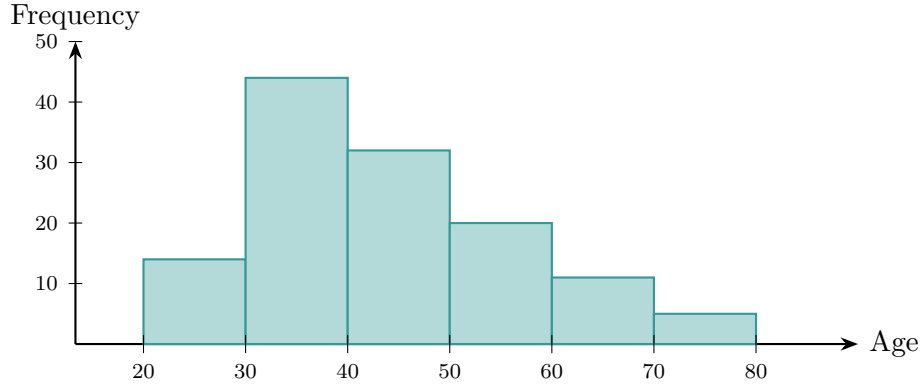


Figure 2.5: Example of a Histogram for a continuous numerical attribute with contiguous bins of constant width.

An incorrect choice of the number of bins profoundly alters the perception of the data:

- An insufficient number of bins causes *oversmoothing*, leveling the distribution and masking skewnesses and multimodality.
- An excessive number of bins causes *undersmoothing*, fragmenting the sample into discontinuous frequencies and emphasizing sampling noise.

For unimodal distributions close to normal and moderate sample sizes n , the recommended number of bins is provided by **Sturges' rule**:

$$k = 1 + \lceil \log_2(n) \rceil \quad (2.5)$$

2.6 Multivariate Visualization Techniques

2.6.1 Scatter Plot and Multi-Dimensional Visualization

The **Scatter Plot** treats pairs of numerical attributes as Cartesian coordinates (x_i, y_i) in the plane. It is the main visual tool for:

- Identifying linear correlations or non-linear functional trends.
- Detecting natural clusters of points (*clusters*).

- Finding multivariate outliers not visible in univariate projections.

Example 2.1 (Analysis of the Anderson/Fisher Iris Dataset). The famous *Iris* dataset (150 samples of flowers belonging to three species: *Setosa*, *Versicolor*, *Virginica*) describes 4 numerical attributes: sepal length and width, petal length and width. The bivariate scatter plot shows that the attributes related to the petal (*Petal Length* vs *Petal Width*) guarantee a clear separation among the species, also allowing the isolation of potential anomalies.

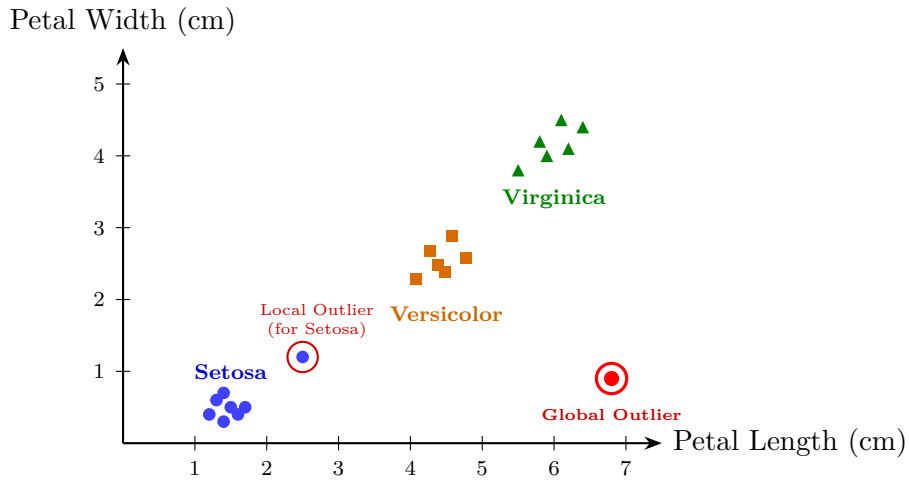


Figure 2.6: Bivariate Scatter Plot with graphical class encoding and visual detection of local and global outliers.

To simultaneously explore all pairs of numerical attributes, a **Scatter Plot Matrix** (SPLOM) is used, which is an $n \times n$ grid where each off-diagonal cell (i, j) displays the scatter plot between attribute i and attribute j , while the main diagonal hosts the histograms or univariate density estimations.

2.6.2 High-Dimensionality Visualization

Parallel Coordinates

When the number of dimensions exceeds 3, the conventional Cartesian representation becomes inapplicable. **Parallel Coordinates** replace the orthogonal axes with a set of equidistant vertical parallel axes, each corresponding to an attribute.

- Each multidimensional object is represented by a **polygonal broken line** that connects the respective scaled values along the vertical axes.

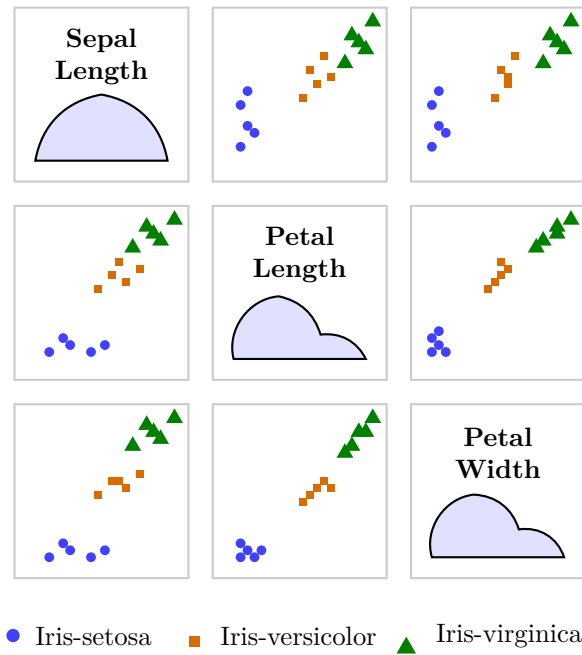


Figure 2.7: Scatter Plot Matrix (SPLOM) of the Iris dataset with the pairwise comparison for all three species and the univariate density estimations along the diagonal.

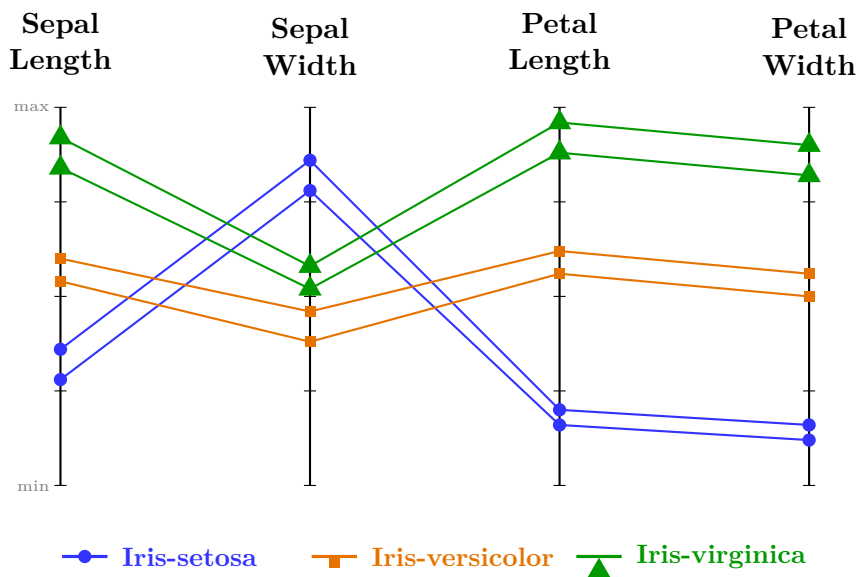


Figure 2.8: Representation of 4-dimensional data using Parallel Coordinates for all three species of the Iris dataset.

- Objects belonging to the same class tend to form bundles of parallel lines or with coordinated trends.
- **Sensitivity to the ordering of the axes:** The visibility of clusters and correlations depends critically on the sequential arrangement of the axes. Two strongly correlated attributes show regular geometric patterns only if positioned as adjacent axes.

Radar Plot and Star Plot

In **Radar Plots**, the axes start from a common center and radiate out in a star shape at constant angular intervals. The value of each attribute defines a vertex along its own radius; joining the vertices yields a polygon for each instance. **Star Plots** apply the same principle by drawing each object in a separate graph, facilitating comparison for individual profiles.

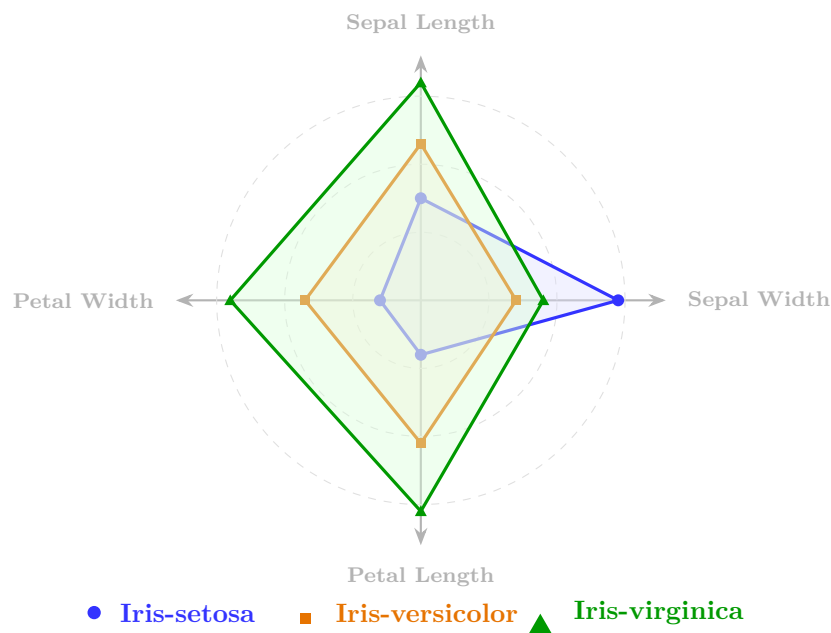


Figure 2.9: Example of a Radar Plot with visual comparison of multivariate profiles for all three Iris species.

2.7 Linear Correlation Analysis

Correlation quantifies the strength and direction of the linear relationship between two continuous numerical attributes X and Y .

Definition 2.2 (Pearson Correlation Coefficient). Given a sample of m pairs of observations (x_i, y_i) , the sample linear correlation coefficient r_{xy} is defined as:

$$r_{xy} = \frac{\sum_{i=1}^m (x_i - \bar{x})(y_i - \bar{y})}{(m-1)s_x s_y} = \frac{\text{Cov}(X, Y)}{s_x s_y} \quad (2.6)$$

where $\bar{x}, \bar{y}$ are the sample means and s_x, s_y the respective standard deviations.

- $r_{xy} \in [-1, +1]$.
- If $|r_{xy}| = 1$, the points (x_i, y_i) lie exactly on a line. If $r_{xy} = +1$ the line has a positive slope; if $r_{xy} = -1$ it has a negative slope.
- Values close to 0 indicate the absence of a **linear** association.
- **Fundamental warning:** $r_{xy} \approx 0$ does not imply statistical independence: deterministic non-linear relationships may exist (for example, parabolic $Y = X^2$) that the Pearson coefficient is unable to detect.

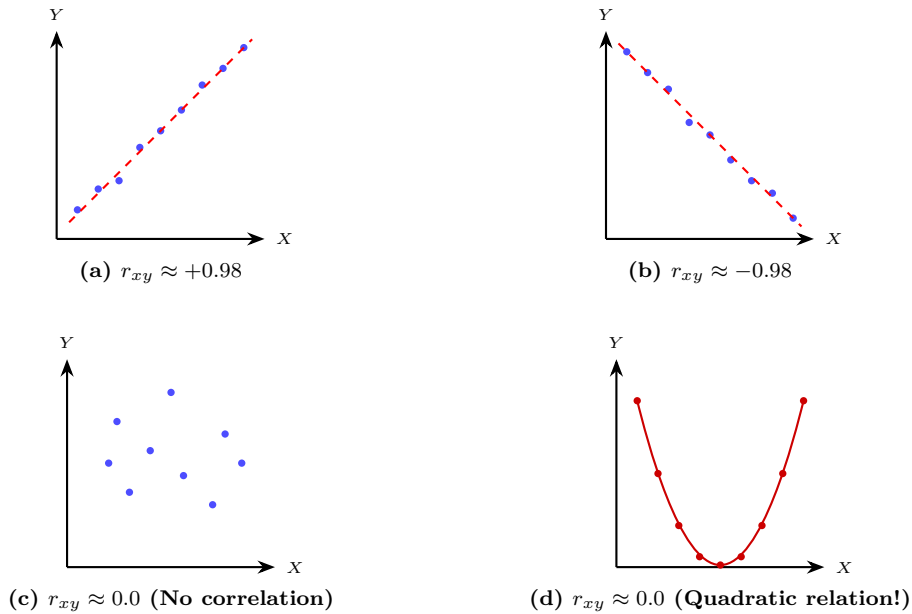


Figure 2.10: Visual interpretation of the Pearson correlation coefficient: (a) positive correlation; (b) negative correlation; (c) absence of linear relation; (d) non-linear deterministic dependence missed by Pearson.

2.8 Handling Outliers and Missing Values

2.8.1 Outliers: Noise vs Knowledge

An **outlier** is an instance with characteristics that deviate markedly from the vast majority of observations:

- **Outliers as noise:** Originating from sensor failures, typos, or transmission inconsistencies. Such anomalies distort measures of central tendency (such as the mean) and predictive models, and must therefore be removed.
- **Outliers as an analytical objective:** Exceptional cases that constitute the true application target (fraudulent bank transactions, unauthorized server access, outbreaks of rare diseases).

Detection methodologies:

- *Single attribute:* Extreme frequency analysis for categorical variables; Box Plot ($1.5 \times \text{IQR}$) and calculation of the z -score ($z = \frac{x-\mu}{\sigma}$) for numerical variables.
- *Multivariate:* Bivariate and trivariate Scatter Plots, projections on principal components (*Principal Component Analysis* - PCA) and clustering algorithms (outliers are the points that fall into residual clusters or remain unassigned).

2.8.2 Missing Values

The causes of failure to record values include:

- Subject's refusal to answer (e.g., information on income or weight).
- Non-applicability of the attribute (e.g., employment income for minors or the unemployed).
- Temporary malfunction of sensing devices.

Pitfall of Dummy Default Values

In many recording systems, a missing value is not saved as NULL or NaN, but rather encoded with a sentinel numeric value such as 0, -1, or 999. If not intercepted in the Data Understanding phase, such encodings are erroneously treated as real values, severely distorting means, standard deviations, and classification models.

2.9 Operational Checklist for Data Understanding

At the end of the exploration phase, the analyst must have completed the following checks:

1. Formal identification of the measurement scale of each attribute.
2. Verification of syntactic accuracy (out-of-domain values) and semantic accuracy.
3. Inspection and removal of duplicate records.
4. Treatment of missing values and elimination of dummy zeros or default encodings.
5. Detection of univariate and multivariate outliers and decision on their treatment (noise to be eliminated vs anomaly to be analyzed).
6. Examination of distributions (skewnesses, multimodalities) via histograms sized according to Sturges' rule.
7. Calculation of the correlation matrix to detect redundant attributes or predictors correlated to the target.

Bibliography

- [1] Usama Fayyad, Gregory Piatetsky-Shapiro, and Padhraic Smyth. From data mining to knowledge discovery in databases. *AI Magazine*, 17(3):37–54, 1996.